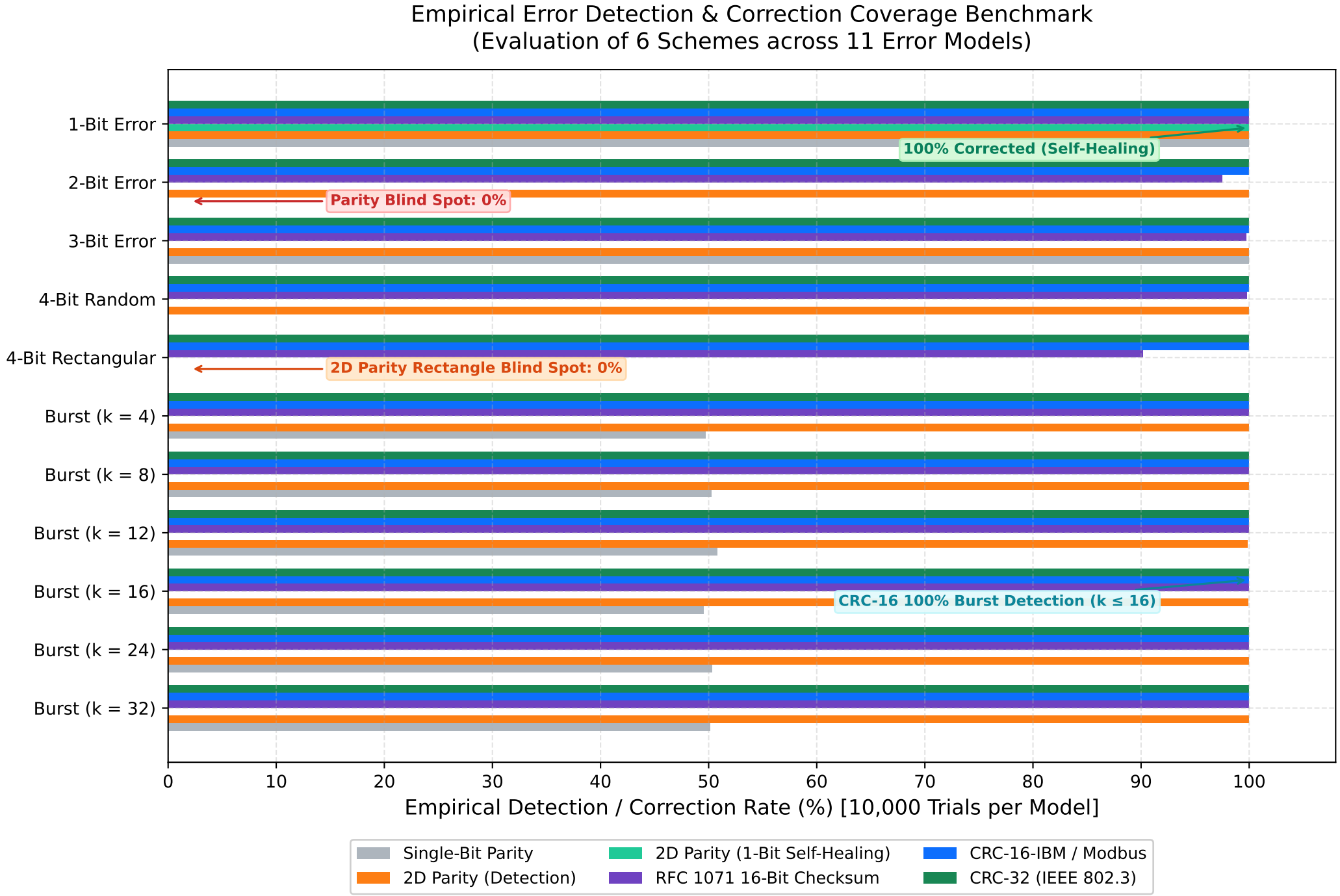
Empirical Error Detection & Correction Coverage Benchmark
(Evaluation of 6 Schemes across 11 Error Models)
1-Bit Error
2-Bit Error
3-Bit Error
4-Bit Random
4-Bit Rectangular
Burst (k = 4)
Burst (k = 8)
Burst (k = 12)
Burst (k = 16)
Burst (k = 24)
Burst (k = 32)
100% Corrected (Self-Healing)
Parity Blind Spot: 0%
2D Parity Rectangle Blind Spot: 0%
CRC-16 100% Burst Detection (k ≤ 16)
0
10
20
30
40
50
60
70
80
90
100
Empirical Detection / Correction Rate (%) [10,000 Trials per Model]
Single-Bit Parity
2D Parity (Detection)
2D Parity (1-Bit Self-Healing)
RFC 1071 16-Bit Checksum
CRC-16-IBM / Modbus
CRC-32 (IEEE 802.3)